

Ghayyas Ahmed

✉ ghayyas1ahmed@gmail.com 🔗 ghayyasahmed.vercel.app ☎ (+92) 3700713489

🌐 linkedin.com/in/ghayyas-ahmed 🏠 github.com/GhayyasAhmed 📍 Karachi, Pakistan

SUMMARY

Full-Stack Web Developer with 4 years of professional frontend experience and hands-on MERN development across production-style applications. Strong in React, Next.js, TypeScript, Node.js, Express.js, MongoDB, REST APIs, authentication, real-time systems, payments, testing, and deployment. Experienced in building scalable frontend architectures and end-to-end applications from UI through backend APIs and cloud deployment.

PROFESSIONAL EXPERIENCE

Front-End Engineer, Kachyng Inc

Feb 2024 – Feb 2026 | *Remote*

- Optimized **Core Web Vitals** for a payment checkout application, reducing **LCP from approximately 2.8s to under 900ms** through **critical rendering path restructuring**, deferred non-critical logic, and improved bundle loading.
- Reduced **JavaScript bundle payload by 200KB+** through **tree-shaking**, native browser APIs, **code splitting**, and **React.lazy/Suspense**, improving production application startup performance.
- Refactored a **2,500-line React module** into a **modular, flow-based architecture** with reusable components and shared UI patterns, improving maintainability across production features.
- Introduced **TypeScript strict mode** across a **React/Vite codebase**, establishing stronger type contracts for API responses, shared state, and reusable components.
- Built **Playwright E2E coverage** for critical payment and iframe workflows across desktop and mobile browsers, contributing to an **80%+ reduction in production incidents**.

Front-End Engineer, Auxin Security

Jun 2022 – Jan 2024 | *Karachi, Pakistan*

- Integrated **8+ backend services** using **React, Redux Toolkit, and RTK Query** into a centralized frontend data layer with **caching and controlled API state management**.
- Implemented **authentication refresh handling** for concurrent API requests and built **RBAC-aware protected workflows** using **Redux Toolkit and custom React hooks**.
- Developed reusable **React and Material UI components** for dashboards, tables, forms, charts, and administrative workflows shared across **multiple SaaS applications**.
- Integrated **Socket.IO real-time events** with Redux, implementing **resilient reconnect and retry handling** for security-monitoring workflows.
- Migrated React applications from **Create React App to Vite with SWC** and supported **Docker, Nginx, and Jenkins-based deployment workflows**.

PROJECTS

Multi-Vendor E-Commerce Platform 🔗, React, Next.js, TypeScript, Node.js, Express.js, MongoDB, Mongoose, Redis, Socket.IO, Stripe, Cloudinary, Docker, Nginx, GitHub Actions, AWS EC2

- **Built a multi-role e-commerce platform** supporting **buyer, seller, and admin workflows**, including **product management**, cart, checkout, orders, seller operations, analytics, and notifications.
- Developed **REST APIs with Express.js, MongoDB/Mongoose, JWT authentication, role-based authorization**, validation, centralized error handling, and secure **cookie-based sessions**.
- Implemented real-time buyer/seller messaging and notifications using **Socket.IO**, with a dedicated socket relay service and Redis-backed communication.
- Integrated **payment processing**, media uploads, transactional email, **Redis session/cache infrastructure**, and protected application workflows using **Stripe, Cloudinary, SMTP, and Redis**.
- Containerized frontend, backend, Redis, and socket-relay services with **Docker Compose**; configured **Nginx** for reverse proxying and **WebSocket routing**, and deployed the stack via **GitHub Actions CI/CD to AWS EC2**.

Learning Management System , Next.js, TypeScript, React, Express.js, MongoDB, Mongoose, Redis, JWT, NextAuth, Google OAuth, Stripe, Cloudinary, VdoCipher

- Built a production-style **Learning Management System** using **Next.js, Express.js, MongoDB, and Redis**, supporting course discovery, enrollment, secure content delivery, and role-based administration.
- Implemented **JWT authentication** with rotating refresh tokens, Redis-backed session management, and role-based authorization to secure student, instructor, and administrator workflows.
- Developed scalable REST APIs for **course management, orders, analytics, user management, and content customization**, following modular Express architecture and centralized error handling.
- Integrated **Stripe payments, Cloudinary media storage, VdoCipher secure video delivery, and Socket.io notifications**, enabling complete purchase, media, and real-time communication workflows.
- Built an admin dashboard with **Redux Toolkit, RTK Query, and MUI DataGrid**, providing course management, user administration, analytics, and responsive data-driven interfaces.

Hotel Booking Platform , React, TypeScript, Node.js, Express.js, MongoDB, Mongoose, Clerk, Stripe, Cloudinary, TanStack Query, Tailwind CSS

- Developed a full-stack hotel booking platform using **React, Express.js, and MongoDB**, enabling end to end workflows for hotel discovery, room reservations, booking management, and owner administration.
- Implemented **Clerk authentication** with user synchronization, protected routes, and webhook based account lifecycle management, ensuring secure authentication across frontend and backend services.
- Integrated **Stripe Checkout** and payment webhooks to automate booking payment confirmation and maintain reliable booking status synchronization.
- Built owner management features with **Express.js, Mongoose, and Cloudinary**, enabling CRUD operations for hotels, rooms, image uploads, booking tracking, and revenue management.
- Designed reusable REST APIs with centralized validation and MongoDB data models, supporting reviews, availability checks, booking cancellation, and scalable backend architecture.

SKILLS

Languages: JavaScript, TypeScript

Frontend: React.js, Next.js, Redux Toolkit, RTK Query, React Hook Form, React Router, React Context API

Backend: Node.js, Express.js, REST APIs, MVC Architecture, Middleware, API Validation, Error Handling

Database & Caching: MongoDB, Mongoose, Redis

Authentication & Security: JWT, HTTP-only Cookies, OAuth, Google OAuth, NextAuth, RBAC, CORS, API Security

Real-Time: Socket.IO, WebSocket Communication, Redis Pub/Sub

UI & Styling: Material UI, Tailwind CSS, Bootstrap, SCSS, Responsive Design, Accessibility

Testing: Playwright, Jest, React Testing Library

Build & Tooling: Vite, Webpack, SWC, npm, Git, GitHub

Deployment & DevOps: Docker, Docker Compose, Nginx, GitHub Actions, AWS EC2, Vercel, Railway

EDUCATION

Bachelor of Science (Computer Science), NED University of Engineering and Technology

2018 – 2022 | *Karachi, Pakistan*